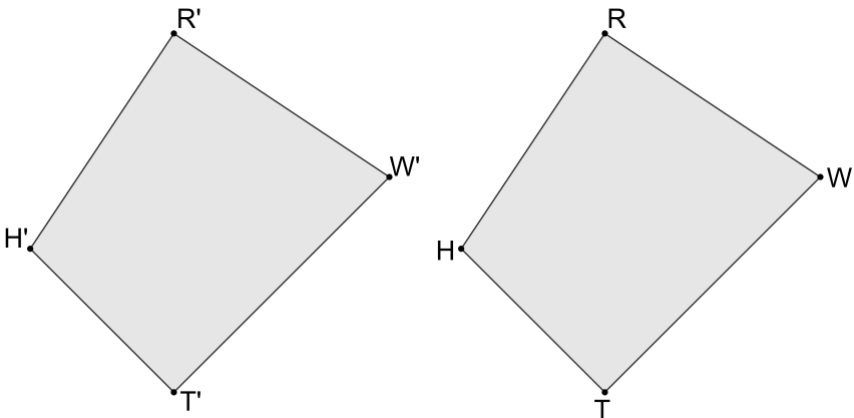
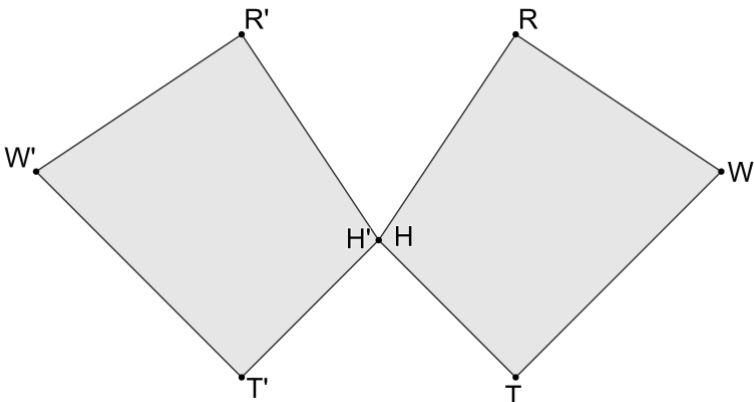
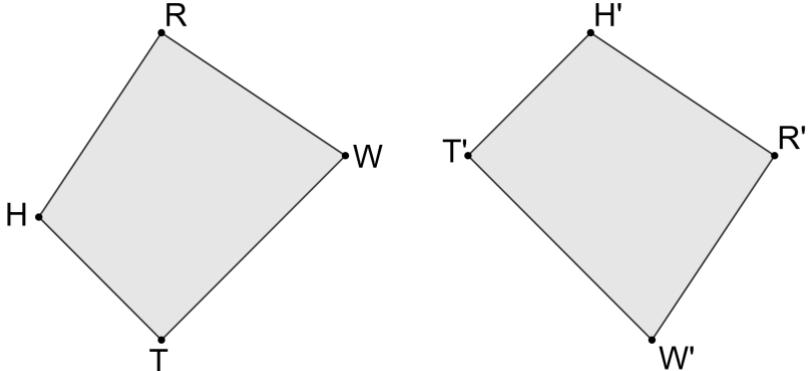
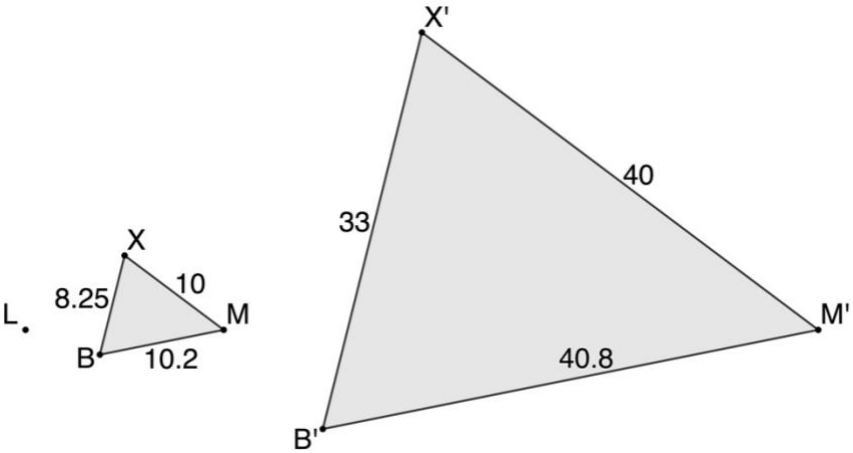


For each movement, quadrilateral $RWTH$ was transformed to create quadrilateral $R'W'T'H'$. Choose the transformation that matches each movement.

	Movement	Transformation
1.		☐ reflection ☐ rotation ☒ translation
2.		☒ reflection ☐ rotation ☐ translation
3.		☐ reflection ☒ rotation ☐ translation

4. Triangle XMB is dilated to create triangle $X'M'B'$, where point L is the center of dilation, as shown.

Determine the scale factor of the dilation.

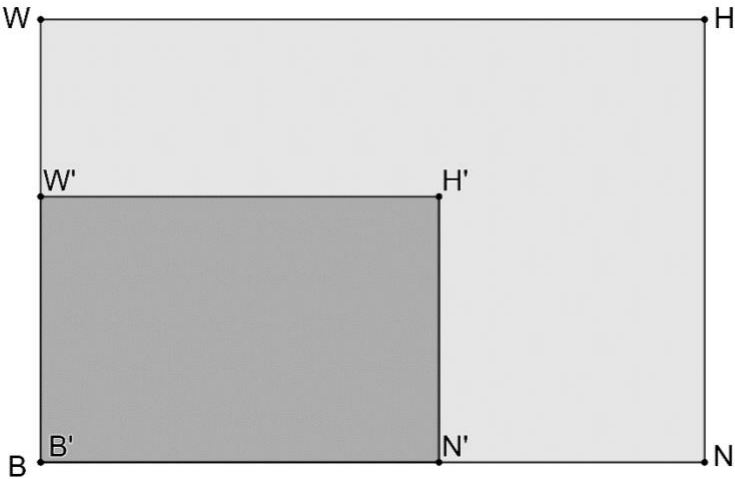


work	
Scale Factor	4

5. Rectangles $WHNB$ and $W'H'N'B'$ are shown, where $W'H'N'B'$ was created using a dilation centered at point B .

Some of the measurements are shown.

$WH = 63.6$ centimeters (cm)
and $NH = 50.4$ cm, and
 $W'B' = 30.24$ cm.



Determine the scale factor of the dilation.

work	
Scale Factor	0.6 or $\frac{3}{5}$

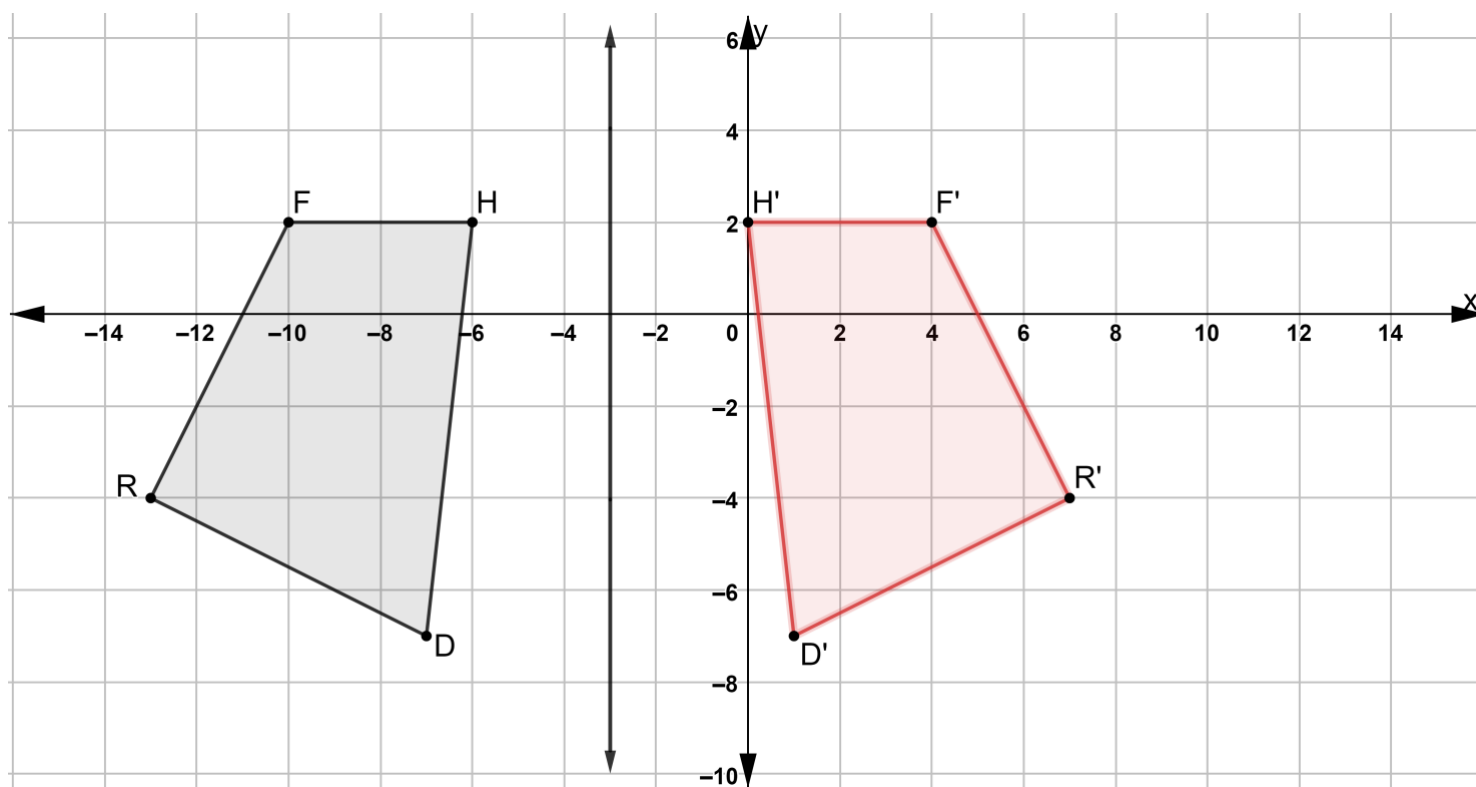
6. The coordinates of the vertices of quadrilateral $FHDR$ are shown in the table.

F	H	D	R
$(-10, 2)$	$(-6, 2)$	$(-7, -7)$	$(-13, -4)$

- a. Quadrilateral $F'H'D'R'$ is created by translating quadrilateral $FHDR$ up 7 units. Complete the table for quadrilateral $F'H'D'R'$.

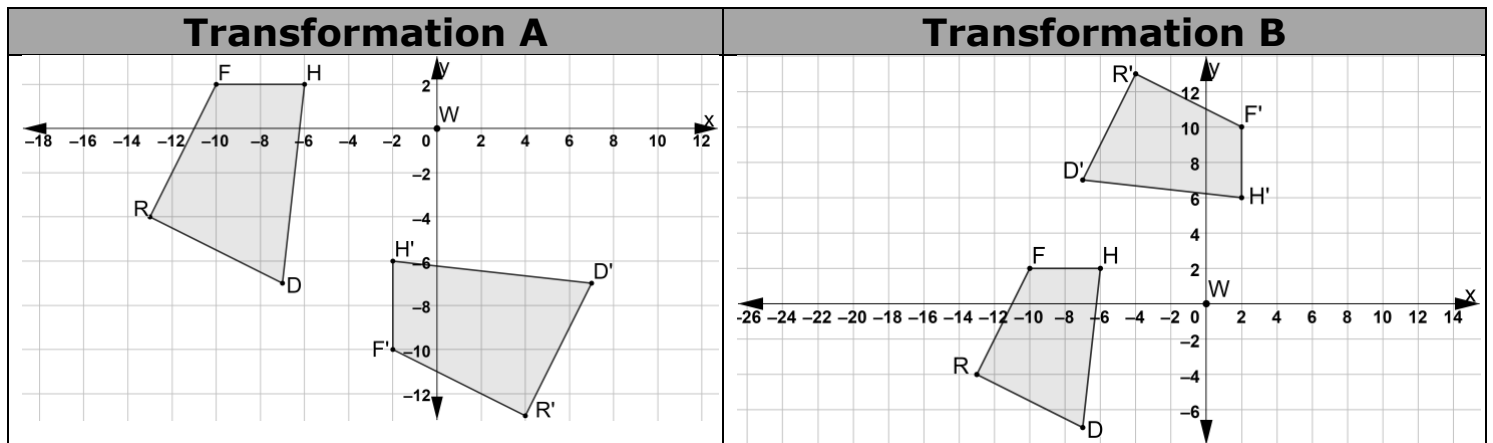
F'	H'	D'	R'
$(-10, 9)$	$(-6, 9)$	$(-7, 0)$	$(-13, 3)$

- b. Quadrilateral $FHDR$ and the line $x = -3$ are shown on the coordinate grid.



Draw quadrilateral $F'H'D'R'$ as a reflection of quadrilateral $FHDR$ across the line $x = -3$.

c. Two transformations of quadrilateral $FHDR$ are shown.



Complete the statements.

Transformation A is rotation of

- ☐ 90° clockwise
- ☐ 180° clockwise
- ☒ 270° clockwise

about point W .

Transformation B is rotation of

- ☐ 90° counterclockwise
- ☐ 180° counterclockwise
- ☒ 270° counterclockwise

about

point W .

7. Complete the statement.

There are two correct ways to complete the statement.

When the mapping of a transformation of a figure shows that the

transformation

- ☒ preserves congruence,
- ☐ has corresponding vertices,
- ☐ is a scaled copy of itself with a scale factor other than 1,

then the

transformation is a

- ☐ slide.
- ☐ dilation.
- ☒ rigid transformation.

When the mapping of a transformation of a figure shows that the

transformation

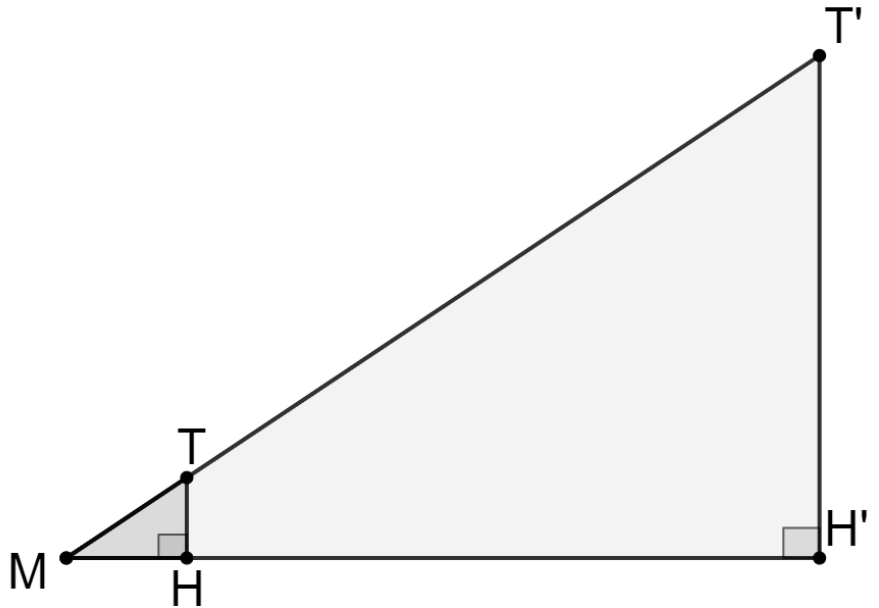
- preserves congruence,
- has corresponding vertices,
- is a scaled copy of itself with a scale factor other than 1,

then the

transformation is a

- slide.
- dilation.
- rigid transformation.

8. A special effects director uses similar triangles to determine the length of shadows created by a spotlight. The director draws right triangle MHT , where $\overline{TH}$ represents the height of the original object. He then dilates triangle MHT using point M as the center of the dilation. Triangles MHT and $M'H'T'$ are shown.



The lengths, in feet (ft), of some of the sides are shown.

MH	TH	MH'
2.5 ft	2 ft	15.625 ft

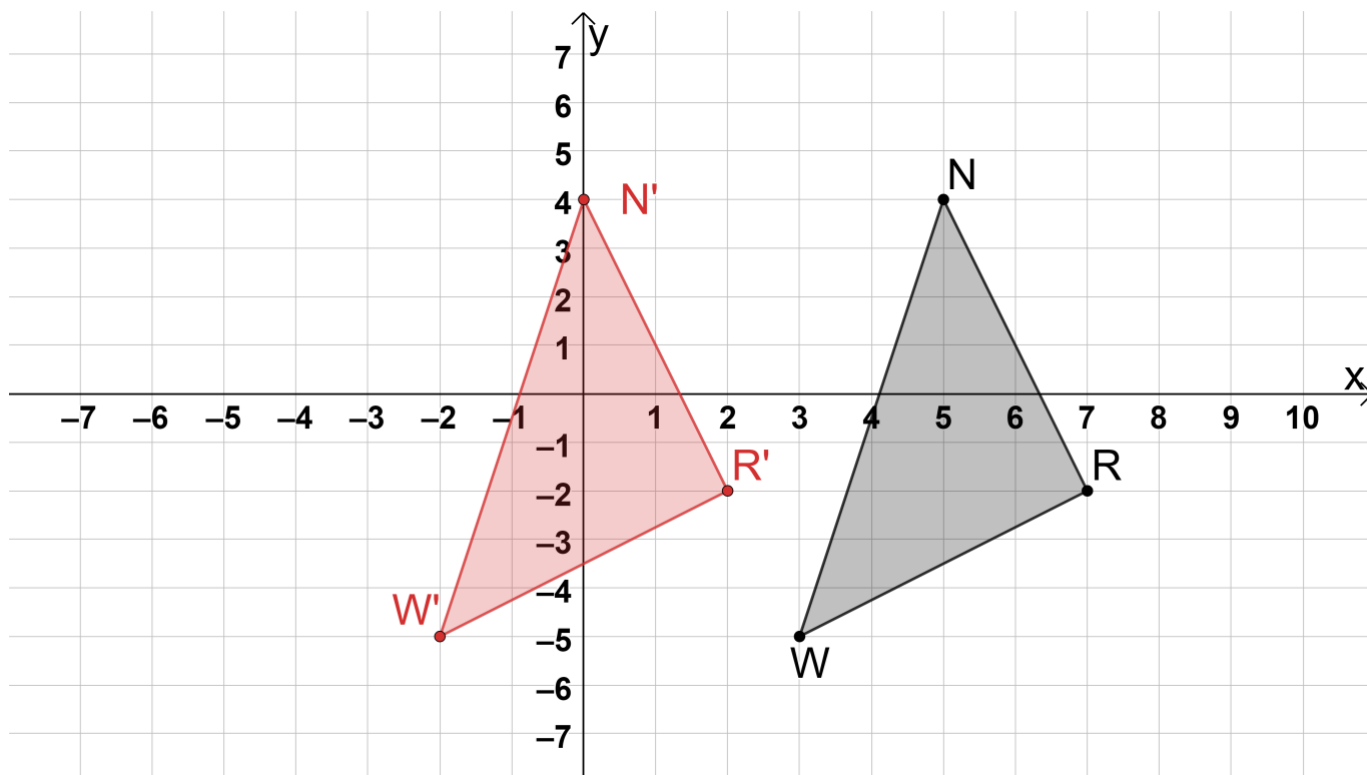
- a. Determine the length, in feet, of $\overline{T'H'}$.

work	
Length	12.5 ft

- b. Determine the length, in feet, of $\overline{MT'}$. Round to the nearest hundredth.

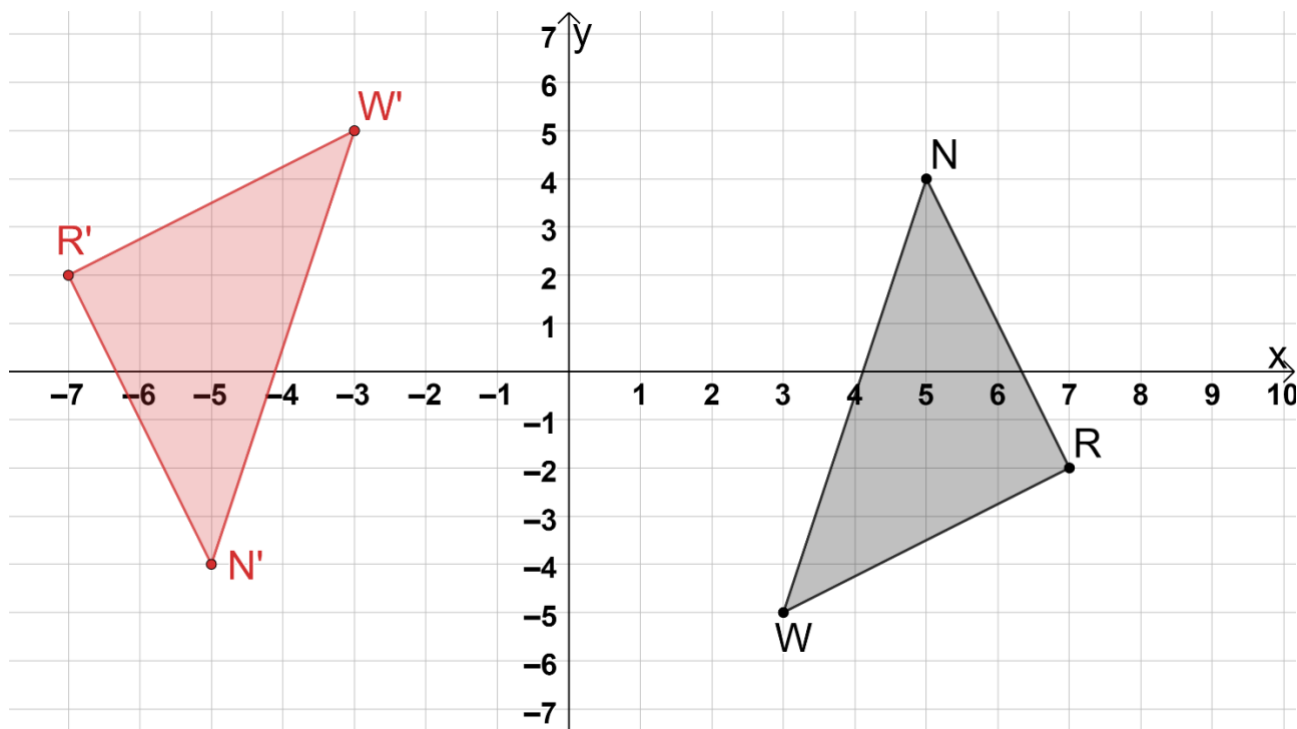
work	
Length	20.01 ft

9. Triangle NWR is shown on the coordinate grid.



Translate $\triangle NWR$ 5 units to the left. Label the transformation $\triangle N'W'R'$.

10. Triangle NWR is shown on the coordinate grid.



Rotate $\triangle NWR$ 180° about the origin. Label the transformation $\triangle N'W'R'$.